Data Analytics Capstone

Retail Sentiment as a Volatility Early-Warning Signal:

A Machine-Learning Study of StockTwits and U.S. Equity Risk

Interim Report

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QM640: Data Analytics Capstone

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Group 12

[Current Term] Term

[Date]

GitHub repository (data and code):

https://github.com/JamesSavages/Retail_Sentiment_and_Market_Volatility

Introduction

Background and Context

Financial markets are driven not only by fundamentals but by the collective mood of the people trading them, which is now broadcast in real time on social platforms and financial news. A substantial literature establishes that this visible sentiment carries information about market behaviour: message-board and microblog activity has been linked to trading volume and volatility (Antweiler & Frank, 2004; Sprenger et al., 2014), and social-media mood to index movements (Bollen et al., 2011). The strongest and most consistent evidence is for volatility and trading volume rather than for returns, which are closer to efficient (Audrino et al., 2020).

For brokerages, robo-advisors and risk-management desks, the operationally valuable question is not what the crowd feels today but whether today's mood signals which stocks will be volatile tomorrow. An accurate early-warning signal for next-day volatility therefore has direct commercial value, informing user risk alerts, position sizing and liquidity decisions.

Measuring sentiment accurately is itself a research problem. Traditional dictionary-based methods have systematically misclassified financial language (Loughran & McDonald, 2011), whereas transformer-based models such as FinBERT read text in context and have improved financial sentiment classification (Araci, 2019). More recently, large language models have shown an emerging capacity for financial reasoning. For example, Lopez-Lira and Tang (2023) found that GPT-4 can predict market reactions to news headlines without task specific training, although returns appear to decay as adoption spreads, further supporting a focus on volatility rather than returns. A further constraint of methodologies used in this field of study is endogeneity. Given that news causes sentiment and price to change at the same time, it is impossible to tell which one is driving the other. To avoid this, this study adopts a predictive design relating sentiment at day *t* to volatility on day *t* + 1, consistent with prior work (Antweiler & Frank, 2004; Audrino et al., 2020).

Problem Statement

Retail investors now generate a continuous public record of market sentiment across social platforms and financial news, and the brokerages and robo-advisors that serve them already surface that stream back to users. What those firms cannot currently do is convert it into a forward-looking risk measure: sentiment is presented descriptively, as a mood gauge of what the crowd feels today, while the operationally useful question is which stocks are likely to move sharply tomorrow. Volatility surprises are where retail users are most exposed and where hedging and liquidity-provision costs fall, so the absence of an early-warning signal is a commercial problem as well as an analytical one. Prior research has largely addressed the question at the market-index level, using lexicon-based sentiment measures, and has reported statistical association rather than economic value, leaving per-firm, cost-aware evidence scarce. This study therefore predicts next-day realized volatility (Y) for eight high-liquidity United States technology equities, with the ticker-day as the unit of analysis, from retail sentiment features (X = relevance-weighted news sentiment and daily article volume as an attention proxy) together with lagged market controls, over January 2023 to December 2024. Success is evaluated by out-of-sample forecast error (root mean squared error, RMSE, and the quasi-likelihood loss, QLIKE) and by the risk-adjusted return (Sharpe ratio) of a sentiment-informed strategy, net of transaction costs.

Purpose of the Study

The study is primarily predictive, forecasting a continuous risk measure, with an explanatory component that quantifies sentiment's marginal information beyond price history, a classification component that predicts next day return direction, and an inferential component that tests whether a sentiment informed strategy significantly improves risk-adjusted return. Its purpose is to establish whether freely available public sentiment provides an operationally useful, cost-aware early-warning signal for equity volatility.

Interim Project Status (Progress Snapshot)

The current status of the project is summarised below.

- **Completed:** Synopsis submitted; data pipeline designed, implemented, and validated end-to-end by a 27-check automated assertion harness; analysis panel assembled (3,840 ticker-days, eight tickers); data cleaning and exploratory data analysis completed; 24-ticker daily price panel collected (Tiingo); news-sentiment collection complete for all eight sentiment tickers; StockTwits labelled corpus accumulating; literature review completed.
- **In progress:** RQ1 sentiment modelling (FinBERT versus the lexicon baseline); specification of the RQ2 volatility models.
- **Pending:** FinBERT sentiment modelling (RQ1); volatility and direction models (RQ2, RQ3); strategy backtest (RQ4); preliminary results and interpretation.

Scope and Objectives

This study is organised around four research questions, each addressing a distinct aspect of the relationship between retail sentiment and equity risk. Each is stated below with its hypotheses and test; all tests are evaluated at the 5% level.

Research Question 1 (RQ1): Sentiment Measurement

How accurately can a transformer model (FinBERT) classify StockTwits message sentiment, benchmarked against users' self-tagged labels and a lexicon baseline of VADER with the Loughran-McDonald dictionary (Araci, 2019; Loughran & McDonald, 2011)? H0: FinBERT's accuracy is no greater than the lexicon's. H1: it is greater. Both classifiers score identical messages, so McNemar's test applies, with a paired bootstrap for balanced accuracy and ranking, since 82% of the corpus is bullish and accuracy alone is uninformative at that base rate.

Research Question 2 (RQ2): Volatility Prediction

Does aggregated daily sentiment at day t predict next-day realized volatility at t + 1, controlling for lagged volatility and trading volume? A heterogeneous autoregressive (HAR) baseline is compared with the same specification augmented by sentiment and attention features, with ticker fixed effects in both, for the reason established in Figure 4. H0: the sentiment coefficients are jointly zero. H1: at least one is non-zero. Testing uses an incremental F-test in-sample and the Diebold-Mariano test (Diebold & Mariano, 1995) on QLIKE losses out-of-sample, following Patton (2011).

Research Question 3 (RQ3): Direction Classification

Do sentiment features improve next-day return-direction prediction beyond a technicals-only baseline? Logistic regression and gradient boosting are each trained on market controls alone, then with sentiment added. H0: the augmented classifier is no more accurate. H1: it is more accurate. Both are evaluated on identical test observations, so McNemar's test applies. Daily direction is close to a coin flip, so a null result is an anticipated and reportable outcome.

Research Question 4 (RQ4): Strategy Evaluation

Does a sentiment-informed trading strategy produce a statistically significant improvement in risk-adjusted return over buy-and-hold, net of transaction costs? The volatility forecast is converted into a transparent exposure rule and back-tested with realistic costs. H0: the strategy's Sharpe ratio is no greater. H1: it is greater. Significance uses the Ledoit-Wolf (2008) robust test with a circular block bootstrap, so that neither normality nor serial independence is assumed.

Literature Survey

Literature Review Approach

Sources were identified through Google Scholar, Scopus, Web of Science, ScienceDirect, SSRN, and arXiv, using Boolean combinations of the study's concept clusters (for example, “investor sentiment” OR “StockTwits” AND “realized volatility” AND “FinBERT” OR “transformer”). Inclusion was limited to peer-reviewed empirical work linking textual sentiment to a measurable market outcome, with field-defining preprints retained where relevant, and priority given to highly cited papers in leading finance and forecasting journals balanced against recency to capture the large-language-model era. Table 1 summarises the most relevant studies.

Table 1

Literature Relevance Matrix

Study	Purpose / Context	Method	Key Findings	Relevance to This Study
Antweiler & Frank (2004)	Message-board sentiment and market activity	Computational-linguistics bullishness index	Messages predict volatility; disagreement predicts volume; return effect small	Anchors the volatility-over-returns focus and the article-count attention feature
Das & Chen (2007)	Extracting sentiment from web small talk	Multi-classifier voting	Sentiment tracks index moves; activity relates to volatility	Early evidence linking posting activity to volatility
Tetlock (2007)	Media tone and the stock market	Media-pessimism factor; VAR	Pessimism predicts price pressure then reversion; extremes predict volume	Frames the endogeneity/simultaneity concern
Bollen et al. (2011)	Social-media mood and the market	Mood dimensions + causality tests	Reported mood relates to index moves; noted as fragile	Motivates a reproducible, firm-level design

Loughran & McDonald (2011)	Sentiment in financial text	Finance-specific dictionaries	General dictionaries misclassify financial tone	Basis for the lexicon baseline in RQ1
Sprenger et al. (2014)	Microblog sentiment and trading	ML sentiment + regression	Sentiment relates to returns; disagreement to volatility	Supports microblog sentiment and attention features
Oliveira et al. (2017)	Microblog data for market prediction	Sentiment indicators + multiple ML	Improves volatility/volume forecasts; weaker for returns	Validates the feature family and volatility target
Renault (2017)	StockTwits investor sentiment	Custom lexicon vs. ML	Sentiment predicts intraday index returns; lexicon competitive	StockTwits-specific precedent and baseline
Behrendt & Schmidt (2018)	Twitter sentiment and intraday volatility	GARCH + intraday sentiment	Adds little to intraday volatility once controlled	Cautionary; motivates conservative effect sizes
Araci (2019)	Transformer sentiment for finance	FinBERT (fine-tuned BERT)	Improves financial sentiment classification	The deep-learning method used in RQ1
Audrino et al. (2020)	Sentiment/attention and volatility	Penalized regression on HAR	Attention and StockTwits volume improve realized-volatility forecasts	Closest precedent; frames the per-ticker, business-evaluated gap
Lopez-Lira & Tang (2023)	LLMs and return predictability	GPT-4 on news headlines	Predicts reactions; edges decay as adoption spreads	Frontier LLM evidence; reinforces volatility focus
Yu et al. (2023)	LLM trading agents	FinMem agent with layered memory	Autonomous agent enhances trading decisions	Grounds the optional agentic briefing extension

Three themes organise this literature and connect it to the present study. First, on outcome selection, the evidence consistently shows that sentiment and attention forecast volatility and trading volume more reliably than returns, which are closer to efficient (Antweiler & Frank, 2004; Oliveira et al., 2017; Audrino et al., 2020); this study therefore targets next-day volatility. Second, on measurement, the field has evolved from finance-specific dictionaries (Loughran & McDonald, 2011) through transformer models (Araci, 2019) to general-purpose large language models (Lopez-Lira & Tang, 2023); RQ1 positions FinBERT against a lexicon baseline within that progression. Third, on the research gap, prior work concentrates at index level, uses lexicon or classical methods, and reports statistical rather than economic value with limited reproducibility; this study addresses that gap by operationalising transformer sentiment as a per-ticker, next-day volatility early-warning signal, validated through a cost-aware backtest and an open pipeline.

Data Description

Data Source(s) and Access

The study draws on three free, publicly accessible sources. Daily split and dividend-adjusted price data are obtained from Tiingo (https://www.tiingo.com); historical per-ticker news sentiment from Alpha Vantage (https://www.alphavantage.co); and recent labelled messages from the StockTwits API (https://api.stocktwits.com). Prices supply the target and market controls, Alpha Vantage supplies the historical sentiment feature for RQ2 to RQ4, and StockTwits supplies the human-labelled corpus for RQ1. During development, an initial price source (Stooq) was found to block automated access and Alpha Vantage's full-history price endpoint was available at a premium price, so Tiingo was adopted after validating access. This deliberate evaluation of data sources strengthens reproducibility and is noted as a practical constraint in the limitations of the final report.

Dataset Overview

The assembled analysis panel is organised as a ticker-day table combining price-derived controls, the next-day realized-volatility target, and news-sentiment features.

- Number of records (rows): 3,840 usable ticker-days (eight sentiment tickers, 480 trading days each) drawn from a 24-ticker, 11,873-row price panel.
- Number of variables (columns): 15 principal variables (see Table 2).
- Time period: 3 January 2023 to 31 December 2024.
- Unit of analysis: the ticker-day.
- Target variable(s): next-day realized volatility (rv_next), estimated from daily open-high-low-close (OHLC) prices via the Parkinson estimator.

Data Dictionary (Mandatory)

Table 2

Data Dictionary (Principal Variables)

Variable Name	Source	Definition	Datatype	Allowed Values/Range	Missing Values Handling	Notes
symbol	All	Stock ticker symbol identifying the equity	string	24 tickers (see config.TICKERS)	None permitted; row invalid without it	Key. Analysis sample restricted to the 8 tickers in NEWS_TICKERS
date	All	Trading date (US/Eastern) of the observation	date	2023-01-03 to 2024-12-31, NYSE trading days	None permitted; row invalid without it	Key. Non-trading days absent by construction, not imputed
close	Tiingo	Split- and dividend-adjusted daily closing price	float	> 0 USD	None observed	Raw. Adjusted series, so corporate actions create no false volatility
log_return	Derived	Daily log return, ln(close_t / close_t-1)	float	approx. -0.5 to +0.5 log points	Null on each ticker's first day; row excluded	Control. Used lagged, never contemporaneously with the target
abs_return	Derived	Absolute daily log return	float	>= 0 log points	Null where log_return is null; row excluded	Control. Correlated with rv; monitored via VIF (Table 5)
log_volume	Derived	Log adjusted share volume, log(1 + volume)	float	> 0 ln(shares)	None observed	Control. log1p so a zero-volume day maps to 0, not minus infinity
rv	Derived	Realized volatility at t from the high-low range (Parkinson)	float	> 0 daily sigma	None observed	Control. Known at the close of t; daily HAR component
rv_lag1	Derived	Realized volatility at t-1	float	> 0 daily sigma	Null on each ticker's first day; row excluded	Control. Retained for continuity with the synopsis specification
rv_w	Derived	Weekly HAR component: mean rv over t-4 to t inclusive	float	> 0 daily sigma	Null for the first 4 days per ticker; rows excluded	Control. Window ends at t because rv_t is observable at that close
rv_m	Derived	Monthly HAR component: mean rv over t-21 to t inclusive	float	> 0 daily sigma	Null for the first 21 days per ticker; rows excluded	Control. Binding on sample size: costs 21 ticker-days per ticker
rv_next	Derived	Next-day realized volatility (t+1)	float	> 0 daily sigma	Null on each ticker's final day; row excluded	TARGET. The only column using information dated after t, by design
news_sent_wmean	Alpha Vantage	Relevance-weighted mean news sentiment on day t	float	-1 (bearish) to +1 (bullish)	Set to 0.0 (neutral) on queried tickers with no articles, flagged by news_missing; null for tickers never queried	Predictor. Weights are Alpha Vantage relevance scores
news_article_count	Alpha Vantage	Articles referencing the ticker on day t; attention proxy	int	>= 0 articles	Set to 0 on queried tickers with no articles (a genuine low-attention observation); null for tickers never queried	Predictor. Zero-article days retained; dropping them truncates attention at 1
news_missing	Derived	1 if the ticker-day had no articles, 0 otherwise	int	{0, 1}	Null for tickers never queried for news	Flag. Separates neutral coverage from absent coverage

Variable Name	Source	Definition	Datatype	Allowed Values/Range	Missing Values Handling	Notes
log_articles	Derived	Log attention proxy, log(1 + news_article_count)	float	>= 0	Null where news_article_count is null	Predictor. Well defined only because zero-article days are retained

Note. The panel contains 15 variables. The StockTwits labelled corpus (id, created_at, symbol, body, sentiment_basic) is used separately for RQ1 sentiment-model validation and is not part of the historical RQ2–RQ4 panel; no user-identifying fields are retained. Generated by src/make_data_dictionary.py and versioned at data/data_dictionary.md.

GitHub Data Availability Statement

The raw and processed datasets and all code are openly available in the project repository:

- Raw data: data/raw/{prices, news, stocktwits}/
- Processed/clean data: data/processed/panel.csv
- Code/notebooks: src/ (collection, feature building, and dictionary scripts)

Screenshots / Evidence

[Insert Figures: repository folder tree, and dataset preview screenshots of the raw folders and the assembled panel.csv, each labelled and referenced in the text (carried over from the synopsis figures).]

Analysis

This section documents how the analysis sample was constructed from the collected data, then reports the exploratory analysis that informed the modelling decisions described in the following section. Every figure and table reported here is generated by src/run_eda.py in the project repository, so all values are reproducible from the committed data.

Data Cleaning

Cleaning is implemented in src/build_features.py and verified by an automated assertion harness, src/validate_panel.py, which recomputes every derived variable from the raw source files and fails if any value disagrees. It runs 27 checks spanning key uniqueness, definitional consistency, news integrity, corporate actions, and the point-in-time test described below. All 27 pass, and the script exits with an error status if any fails, so a corrupted panel cannot silently reach the modelling stage.

The collected price panel contains 11,873 ticker-days across 24 technology equities, 175 fewer than the 12,048 implied by 24 tickers over 502 trading days because Arm Holdings (ARM) listed in September 2023. Restricting to the eight tickers with news sentiment leaves 4,016 ticker-days; removing the final trading day of each series, where the target is undefined, and the first 21 days, where the monthly heterogeneous autoregressive (HAR) component has insufficient history, yields a balanced analysis sample of 3,840 ticker-days, exactly 480 per ticker. Table 3 records each step.

Table 3

Data Cleaning Log

Issue	Variables Affected	Detection Method	Treatment Applied	Records Affected	Rationale
Duplicate ticker-days	symbol, date	Key-uniqueness assertion in validate_panel.py	None found; no action required	0 (0.00%)	Data quality
Tickers outside the sentiment universe	news_sent_wmean, news_article_count	Free-tier quota of 25 requests per day	Excluded from the analysis sample; retained in the price panel as market context	7,857 (66.18%)	Quota constraint, not a data defect
Target undefined at series end	rv_next	Null check after the t to t+1 shift	Excluded, one ticker-day per ticker	8 (0.07%)	Next-day volatility does not exist for the final observation
HAR burn-in window	rv_m	22-day rolling mean requires 22 prior observations	Excluded, 21 ticker-days per ticker	168 (1.42%)	Monthly HAR component undefined until the window fills

No news articles on a trading day	news_sent_wmean, news_article_count	Gap between the news files and the trading calendar	Retained as zero-attention observations: article count 0, sentiment 0 (neutral), indicator news_missing = 1	945 (7.96%)	A zero-article day is a genuine low-attention observation, not a missing measurement
Extreme one-day price moves	close, rv	Screen for absolute close-to-close moves above 30%, cross-checked against split ratios and the volatility path	Both flagged moves verified as post-earnings repricings and retained; no winsorising or trimming applied	2 (0.02%)	Extreme volatility is the phenomenon under study, not noise to remove
User-identifying fields	user_id, user_followers	Privacy review of the StockTwits collector	Removed at collection and stripped from 23 existing files	23 files	Ethics: no individual user is profiled or re-published

Note. Percentages are of the 11,873-row collected price panel. Exclusions are sequential, leaving an analysis sample of 3,840 ticker-days. Counts are produced by src/run_eda.py and written to docs/tables/eda_cleaning_log.csv.

Two treatment decisions warrant explanation. The first concerns the 945 ticker-days, 24.6% of the analysis sample, on which no article mentioned the ticker. An earlier version of the pipeline discarded these rows because the join produced no match. That was incorrect: a day with no coverage is a genuine low-attention observation, not a failed measurement, and discarding such days truncates the attention variable at one article, removing the low-attention contrast the study depends upon. The loss would also have fallen unevenly, with Palantir forfeiting 73% of its history against Microsoft's 4%. These days are retained with an article count of zero, a neutral sentiment score, and an indicator distinguishing neutral coverage from absent coverage. The correction raised the analysis sample from 3,026 to 3,840 ticker-days and balanced the panel at 480 days per ticker.

This treatment is applied only to the eight tickers actually queried. For the remaining sixteen no request was issued, so their news fields remain null; recording a zero would fabricate an observation from an absence of collection, and the validation harness asserts none exists.

The second decision concerns outliers. No trimming or winsorising was applied. A screen for absolute close-to-close moves above 30% flagged two observations: ARM on 8 February 2024 (+47.9%) and PLTR on 6 February 2024 (+30.8%). An unadjusted split produces a price ratio sitting on a simple fraction, and neither ratio, 1.479 and 1.308, does so; a split also leaves volatility unchanged, whereas both show volatility rising over several sessions, ARM from 0.030 to 0.179. Both are post-earnings repricings. Since the study forecasts exactly such episodes, removing them would remove the signal of interest.

Because the study makes a forecasting claim, cleaning must guarantee that no predictor contains information dated after day t. This is verified rather than asserted: for a random sample of ticker-days the harness rebuilds every predictor from raw records dated on or before t and requires an exact match, and across all 11,873 ticker-days the largest discrepancy is of the order of ten to the power of minus sixteen. Realized volatility at t is computed from that day's high and low, so it is known at the close and is admissible as a predictor of t + 1; the HAR windows accordingly end at t rather than being shifted back.

Exploratory Data Analysis

Table 4 summarises the analysis sample. The target and its lagged components share a common scale, so differences between them are informative: averaging volatility over longer windows progressively removes the right tail, with skewness falling from 1.81 for the daily series to 0.62 for the monthly average.

Table 4

Descriptive Statistics for the Analysis Sample

Variable	N	M	Mdn	SD	IQR	Min	Max	Skew	Kurtosis
rv_next (target)	3,840	0.0184	0.0158	0.0107	0.0124	0.0032	0.1003	1.81	5.42
rv (daily HAR)	3,840	0.0185	0.0158	0.0108	0.0124	0.0032	0.1003	1.81	5.35
rv_w (weekly HAR)	3,840	0.0185	0.0170	0.0085	0.0117	0.0045	0.0583	1.07	1.43

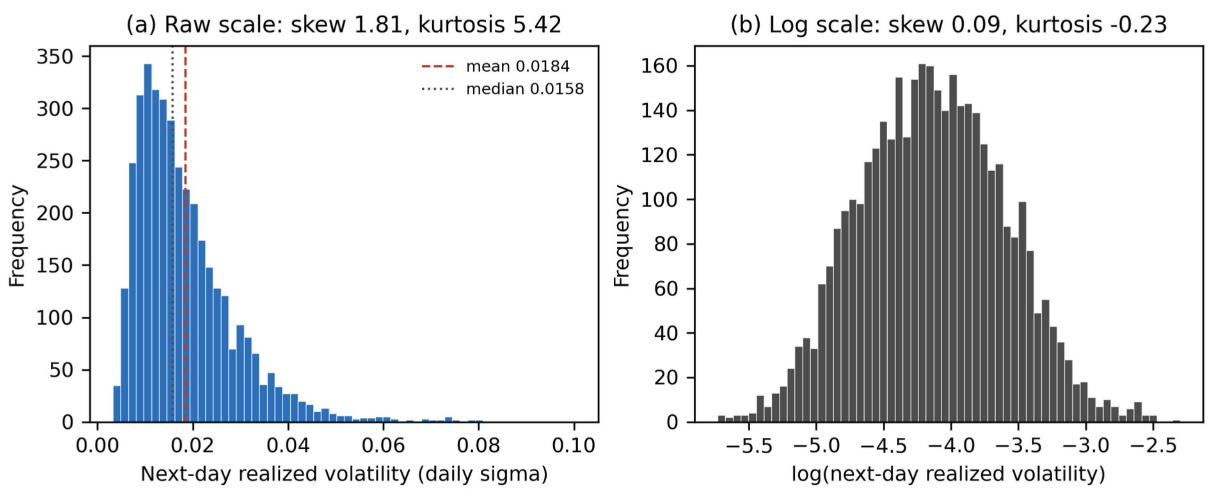
rv_m (monthly HAR)	3,840	0.0186	0.0178	0.0074	0.0115	0.0070	0.0467	0.62	-0.11
news_sent_wmean	3,840	0.1365	0.1320	0.1774	0.2677	-0.6165	0.8746	-0.12	0.57
news_article_count	3,840	2.20	2.00	2.32	2.00	0	26	2.30	12.10
log_articles	3,840	0.9354	1.0986	0.6770	0.6931	0.0000	3.2958	0.08	-0.82
log_return	3,840	0.0022	0.0013	0.0275	0.0252	-0.1638	0.2685	1.08	10.54
log_volume	3,840	17.85	17.74	0.99	1.17	15.37	21.16	0.63	0.10

Note. Realized volatility is expressed in daily sigma units. Sentiment is relevance-weighted and bounded at -1 and +1. Values are written to docs/tables/eda_descriptives.csv.

Figure 1 shows why the target is modelled on a logarithmic scale. Raw next-day realized volatility is strongly right-skewed (skewness 1.81, kurtosis 5.42): most days are quiet and a few extreme. Logging reduces these to 0.09 and -0.23, close to normal. Two decisions follow: the regression target is the logarithm of next-day volatility, so estimates are not dominated by a handful of earnings days, and accuracy is reported using QLIKE alongside RMSE, because squared-error loss on a fat-tailed target rewards fitting the extreme tail at the expense of the rest.

Figure 1

Distribution of Next-Day Realized Volatility, Raw and Log Scale

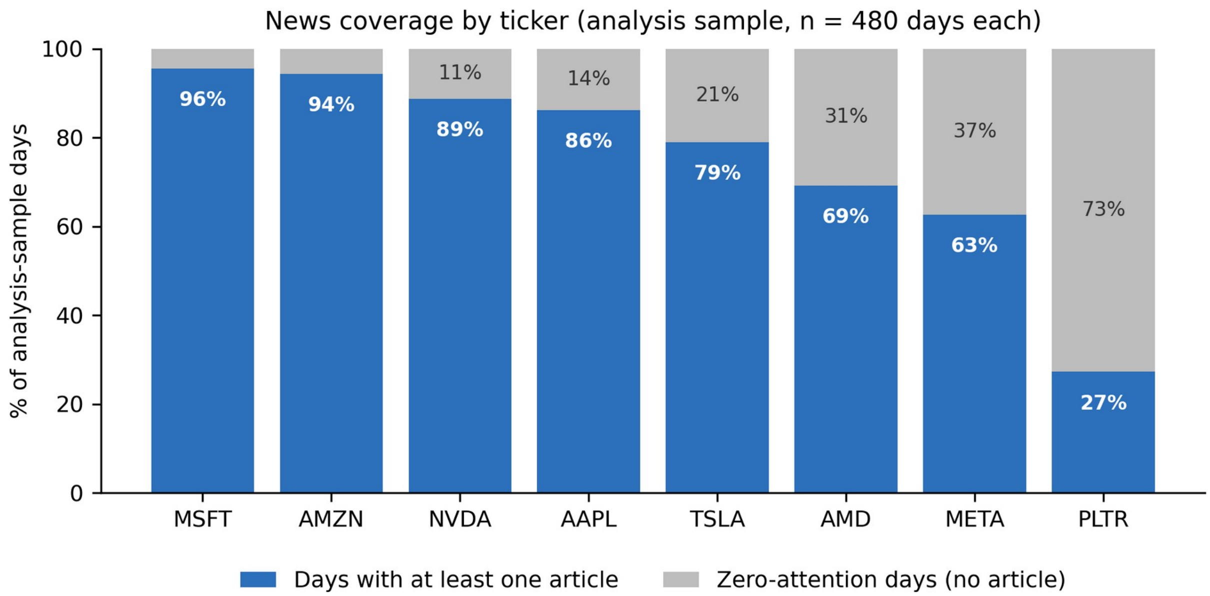


Note. Panel (a) shows the raw target and panel (b) its natural logarithm, each with 60 bins.

Figure 2 shows that news coverage is highly uneven. Microsoft is covered on 96% of trading days and Amazon on 94%, whereas Palantir is covered on only 27%. Coverage is therefore a characteristic that differs systematically between firms, which is the principal justification for retaining zero-attention days and including the news_missing indicator. It also anticipates the confounding problem identified in Figure 5.

Figure 2

News Coverage by Ticker Within the Analysis Sample

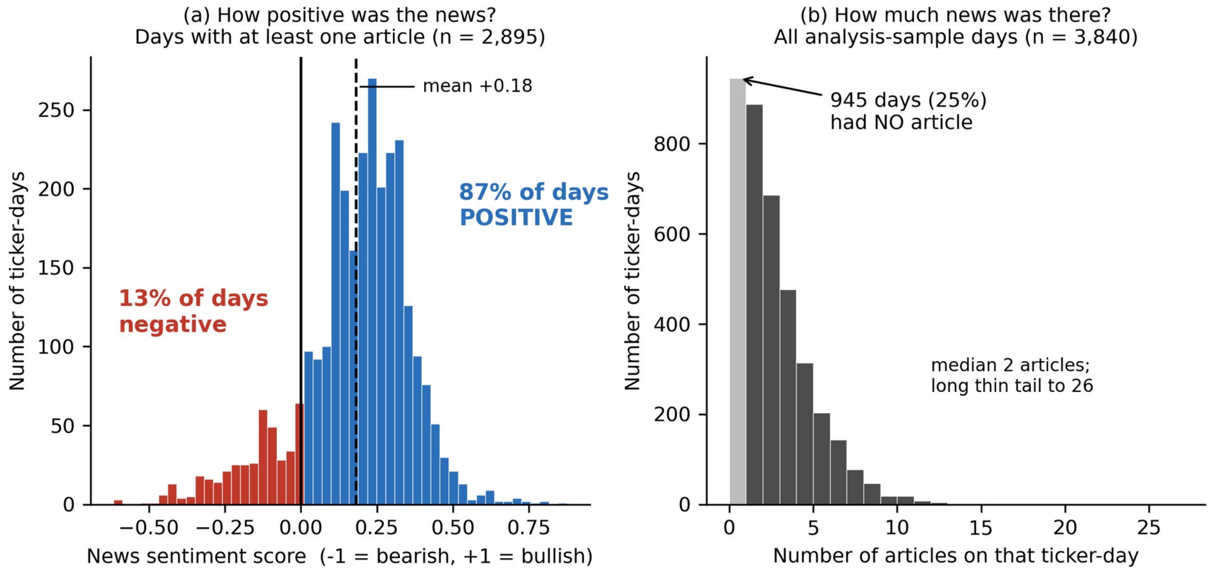


Note. Each ticker contributes exactly 480 ticker-days. Shaded portions are days on which no article mentioned the ticker.

Figure 3 reveals an asymmetry with direct consequences. Relevance-weighted sentiment averages +0.181 and is positive on 87% of days with news, so the measure carries far less negative than positive variation. This compresses the range over which any sentiment effect could be detected and is recorded as a limitation. Panel (b) shows attention is concentrated at low values, median two articles and maximum 26, which is why it enters the models in logarithmic form.

Figure 3

Distribution of News Sentiment and Attention

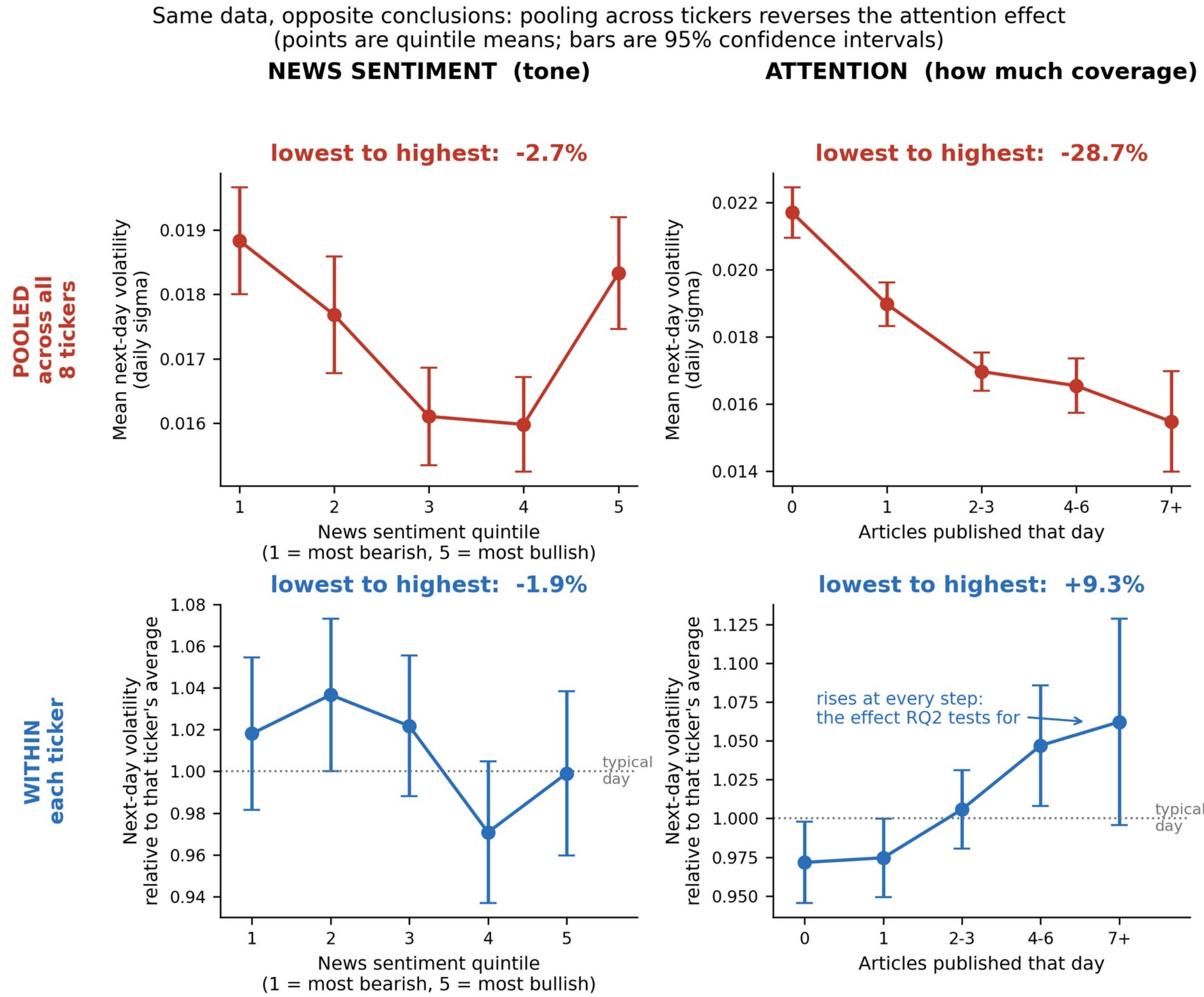


Note. Panel (a) is restricted to days with at least one article; panel (b) includes all analysis-sample days.

Figure 4 contains the most consequential finding of the exploratory analysis. Pooled across tickers, moving from the lowest to the highest attention band is associated with 28.7% lower next-day volatility, contradicting the established result that attention and volatility move together (Audrino et al., 2020). Comparing each ticker against its own mean reverses the sign: days with seven or more articles carry 9.3% higher next-day volatility than days with none, rising monotonically across all five bands. Sentiment shows no consistent pattern in either specification, at -2.7% pooled and -1.9% within ticker. The pooled estimate is an artefact, for the reason set out in Figure 5; the models addressing RQ2 must therefore include ticker fixed effects, and attention rather than tone emerges as the stronger candidate predictor.

Figure 4

Next-Day Volatility by Predictor Group, Pooled and Within Ticker

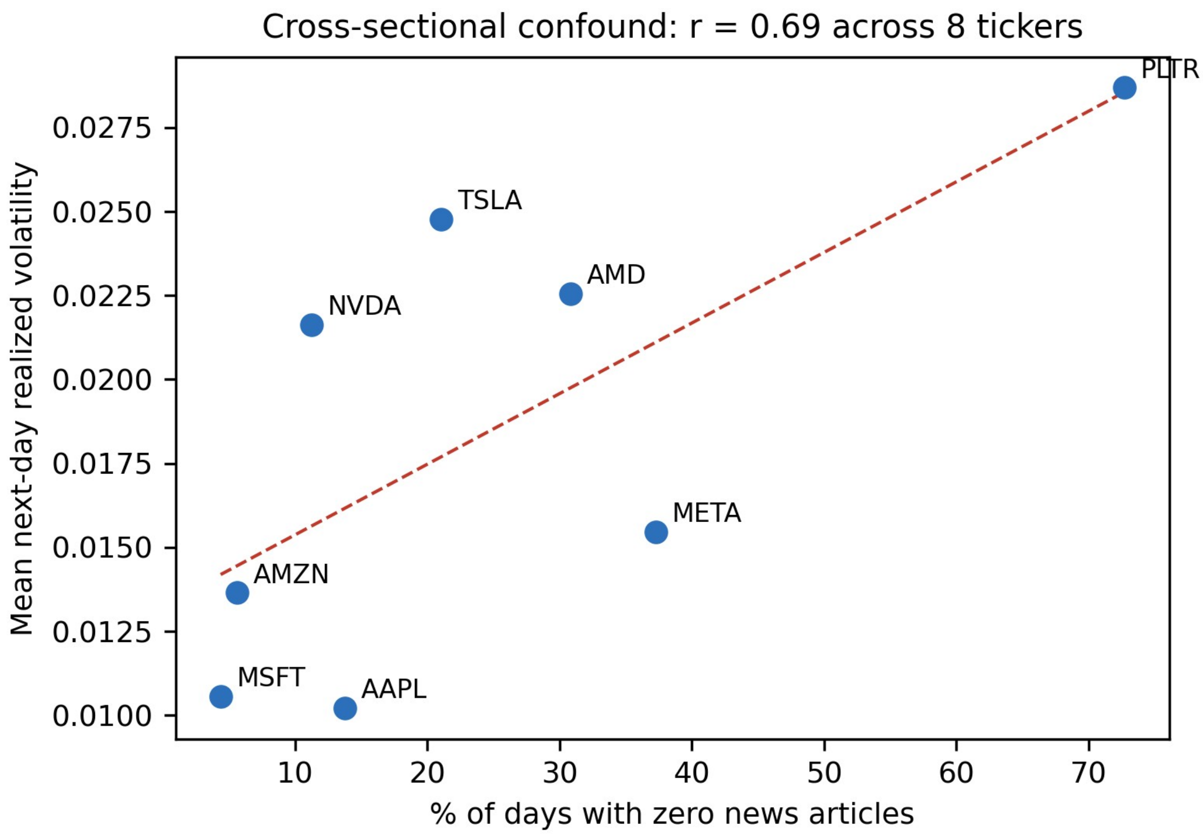


Note. Upper panels pool all tickers; lower panels express the target relative to each ticker's own mean, so 1.00 denotes a typical day for that stock. Sentiment is grouped into quintiles; attention is grouped by article count rather than quintiles, because a quarter of ticker-days have exactly zero articles and quantile boundaries would either collapse or split identical observations. Error bars are 95% confidence intervals.

Figure 5 explains the sign reversal. Across the eight tickers, the proportion of zero-attention days correlates 0.69 with average volatility: Palantir has the least coverage and the highest volatility, Microsoft the most coverage and among the lowest. In pooled data, low attention therefore functions largely as a label for which stock is being observed rather than as information about what is about to happen to it. This is a Simpson's paradox, in which a relationship holding within each group reverses when the groups are combined. Because RQ2 asks whether a change in a ticker's own attention predicts a change in its own volatility, only the within-ticker comparison is informative.

Figure 5

Cross-Sectional Confounding Between Coverage and Baseline Volatility

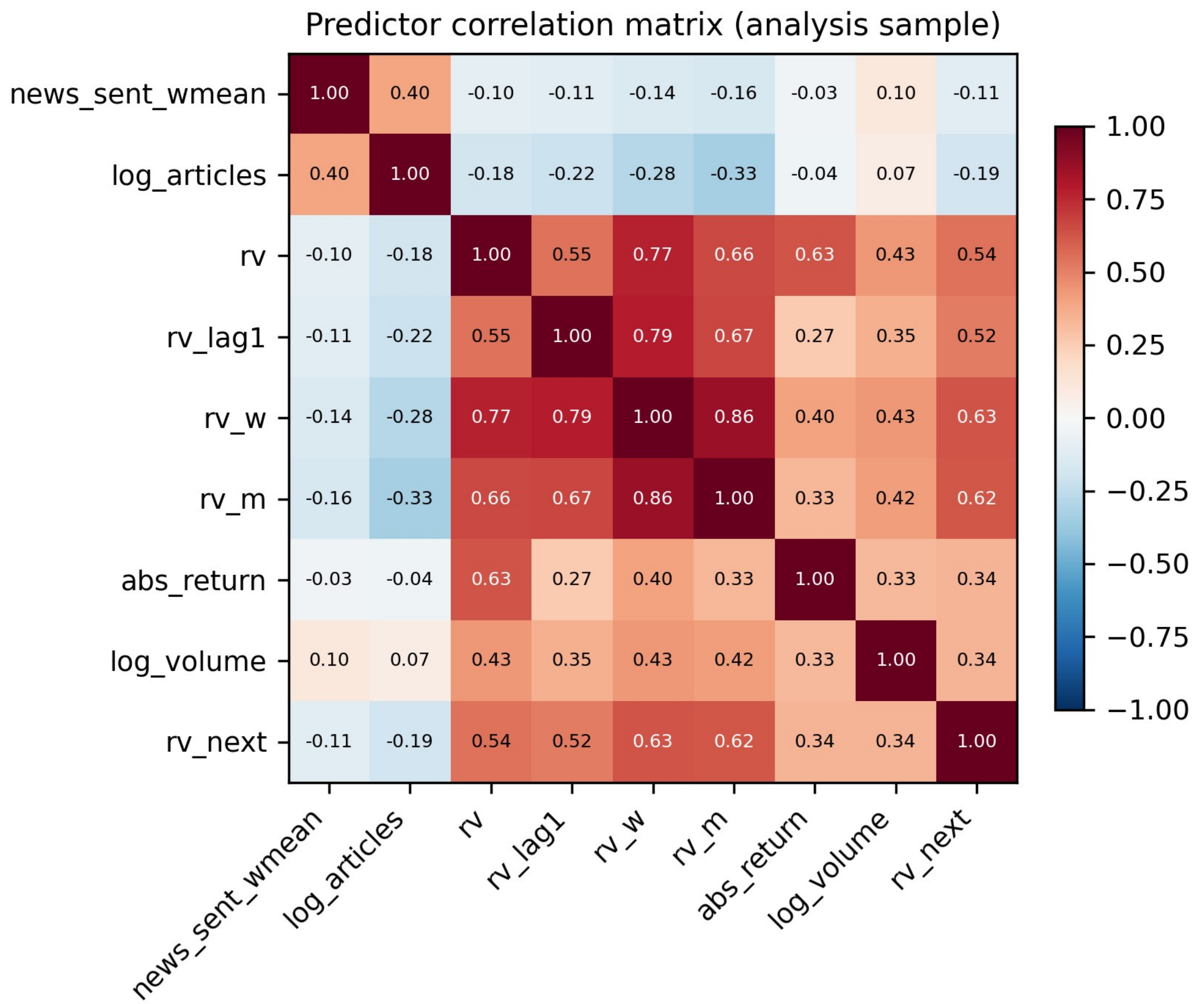


Note. Each point is one of the eight tickers in the analysis sample; the dashed line is an ordinary least squares fit.

Figure 6 confirms that lagged volatility dominates the predictor set: the weekly HAR component correlates 0.63 with the target, news sentiment only -0.11 and attention -0.15. This determines how RQ2 must be evaluated. Sentiment cannot be judged by its raw association with next-day volatility, which is small, but by whether it adds explanatory power once the HAR components are in the model. The hypothesis test for RQ2 is therefore an incremental one.

Figure 6

Correlation Matrix of Predictors and Target



Note. Pearson correlations computed on the 3,840-observation analysis sample.

Multicollinearity was assessed using variance inflation factors (Table 5). Step 4 identifies a factor above 10 as severe; none approaches it. The weekly HAR component is highest at 7.91, expected since it is a moving average of the daily series. The two sentiment variables are close to orthogonal to the market controls, at 1.36 and 1.20, so whatever incremental contribution they make cannot be an artefact of overlap with lagged volatility. The full control set is retained unchanged.

Table 5

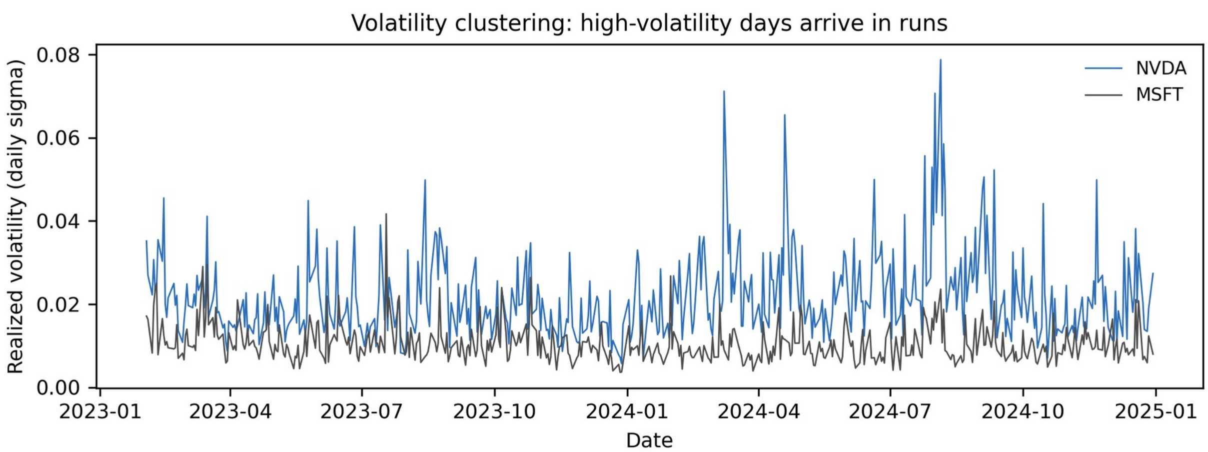
Variance Inflation Factors for the Full Predictor Set

Predictor	VIF	Assessment
rv_w (weekly HAR component)	7.91	Moderate (5-10)
rv_m (monthly HAR component)	4.20	Acceptable (below 5)
rv (daily HAR component)	3.63	Acceptable (below 5)
rv_lag1	2.71	Acceptable (below 5)
abs_return	1.74	Acceptable (below 5)
log_volume	1.40	Acceptable (below 5)
log_articles	1.36	Acceptable (below 5)
news_sent_wmean	1.20	Acceptable (below 5)

Note. Computed on the analysis sample and written to docs/tables/eda_vif.csv. No predictor exceeds the severe threshold of 10.

Figure 7

Realized Volatility Over Time for NVDA and MSFT



Note. Daily realized volatility across the 2023 to 2024 study window for two contrasting large-capitalisation tickers.

Volatility arrives in runs rather than independently from day to day: quiet periods persist and turbulent periods persist. This is why a forecasting model can achieve substantial accuracy from lagged volatility alone, and why the benchmark for RQ2 is a HAR baseline rather than a naive mean. It sets a demanding standard for the sentiment features, which must improve on a baseline that is already informative.

Table 6

Exploratory Data Analysis Insight Summary

Figure/Table Reference	What it Shows	Key Insight	Why it Matters (Link to RQ/Objective)	Decision/Next Step
Figure 1	Distribution of the target, raw and logged	Right-skewed (1.81), fat-tailed; near-normal once logged	RQ2 estimates would be dominated by a few extreme days	Model the log target; report QLIKE alongside RMSE
Figure 2	News coverage by ticker	Coverage ranges from 27% (PLTR) to 96% (MSFT)	Determines whether zero-attention days can be dropped	Retain zero-attention days; include news_missing
Figure 3	Sentiment and attention distributions	Sentiment positive on 87% of days; attention concentrated at low counts	Limited negative variation caps the detectable sentiment effect	Use log attention; record the asymmetry as a limitation
Figure 4	Group means, pooled versus within ticker	Attention effect reverses sign: -28.7% pooled, +9.3% within	A pooled model would report the wrong sign for RQ2	Include ticker fixed effects in the RQ2 and RQ3 models
Figure 5	Coverage against baseline volatility across tickers	Correlation of 0.69 produces a Simpson's paradox	Identifies the cause of the reversal in Figure 4	Report within-ticker results as primary, pooled as contrast
Figure 6	Predictor correlation matrix	rv_w correlates 0.63 with the target; sentiment only -0.11	Sentiment cannot be assessed on raw correlation alone	Test sentiment by incremental contribution over HAR
Table 5	Variance inflation factors	Maximum VIF 7.91; sentiment features at 1.36 and 1.20	Confirms the coefficients will be interpretable	Retain the full control set unchanged
Figure 7	Realized volatility over time	Pronounced volatility clustering	Justifies the HAR benchmark used for RQ2	Benchmark against HAR rather than a naive mean

Note. Each row states the interpretation drawn from the exhibit and the design decision that follows from it.

Sample Size Computation

Each research question is sized by the method matching its statistical form: a confidence interval where a single quantity is estimated, and power analysis where an effect is tested. Effect sizes are set conservatively, because the literature consistently reports that sentiment effects on market outcomes are statistically real but economically small (Antweiler & Frank, 2004; Audrino et al., 2020). Table 7 states the method, parameters and substitution for each question.

Table 7

Minimum Sample Size by Research Question

RQ	Method	Formula	Parameters	Substitution	Minimum

					N
RQ1	Confidence interval for a proportion	n = Z^2 p(1-p) / e^2	Z = 1.96 (95%); p = 0.50; e = 0.05	(1.96)^2 (0.50)(0.50) / (0.05)^2	385 labelled messages
RQ2	Power analysis, multiple regression (Cohen, 1988)	N = L / f^2 + u + 1	alpha = .05; power = .80; f^2 = 0.02 (small); u = 3; L = 10.90	10.90 / 0.02 + 3 + 1	549 ticker-days
RQ3	Power analysis, one-sample proportion	n = [Z(1-a) sqrt(p0 q0) + Z(1-b) sqrt(p1 q1)]^2 / (p1 - p0)^2	Z(1-a) = 1.645; Z(1-b) = 0.8416; p0 = 0.52; p1 = 0.57	[1.645 sqrt(.2496) + .8416 sqrt(.2451)]^2 / (.05)^2	614 ticker-days
RQ4	Power analysis, one-sample t test on the Sharpe ratio	n = [Z(1-a) + Z(1-b)]^2 / d^2	Z(1-a) = 1.645; Z(1-b) = 0.8416; d = 0.063 (annual Sharpe 1.0 / sqrt(252))	(1.645 + 0.8416)^2 / (0.063)^2	1,558 trading days

Note. Effect sizes follow Cohen's (1988) benchmarks. For RQ2 the tested predictors are relevance-weighted sentiment, log attention and the zero-attention indicator, so u = 3 and the corresponding tabulated constant is L = 10.90.

Because all four analyses draw on the same data, the study adopts the largest of the four minima. Three are met comfortably: the labelled corpus contains 4,222 messages against the 385 required for RQ1, and the panel contains 3,840 ticker-days against the 549 and 614 required for RQ2 and RQ3. RQ4 is the binding and unmet constraint. Its requirement of 1,558 observations is expressed in trading days and the window contains only 500. Pooling the eight tickers does not resolve this, because their daily returns are strongly cross-correlated, so the effective number of independent observations is far below the row count. RQ4 is therefore under-powered by design, its test uses a block bootstrap to respect that dependence, and a null result must be read as inconclusive rather than as evidence of no effect.

Meeting a minimum sample size is necessary but not sufficient, because the ticker-days are not independent of one another. Table 8 therefore restates the available sample on three bases and reports the smallest effect RQ2 could detect on each. Even on the most conservative basis the design retains power to detect a small effect in Cohen's (1988) terms, which is the magnitude the literature leads the study to expect.

Table 8

Effective Sample Size and Minimum Detectable Effect for RQ2

Basis	N	Rationale	Detectable f^2	Interpretation
Pooled ticker-days	3,840	Raw row count; assumes every observation is independent	0.003	Optimistic; overstates precision
Correlation-adjusted	1,093	8 / (1 + 7 x 0.359) = 2.28 effective series x 480 dates	0.010	Primary basis reported in this study
Distinct trading dates	480	Treats each date as one cluster, ignoring cross-sectional information	0.023	Most conservative; still near the small-effect threshold

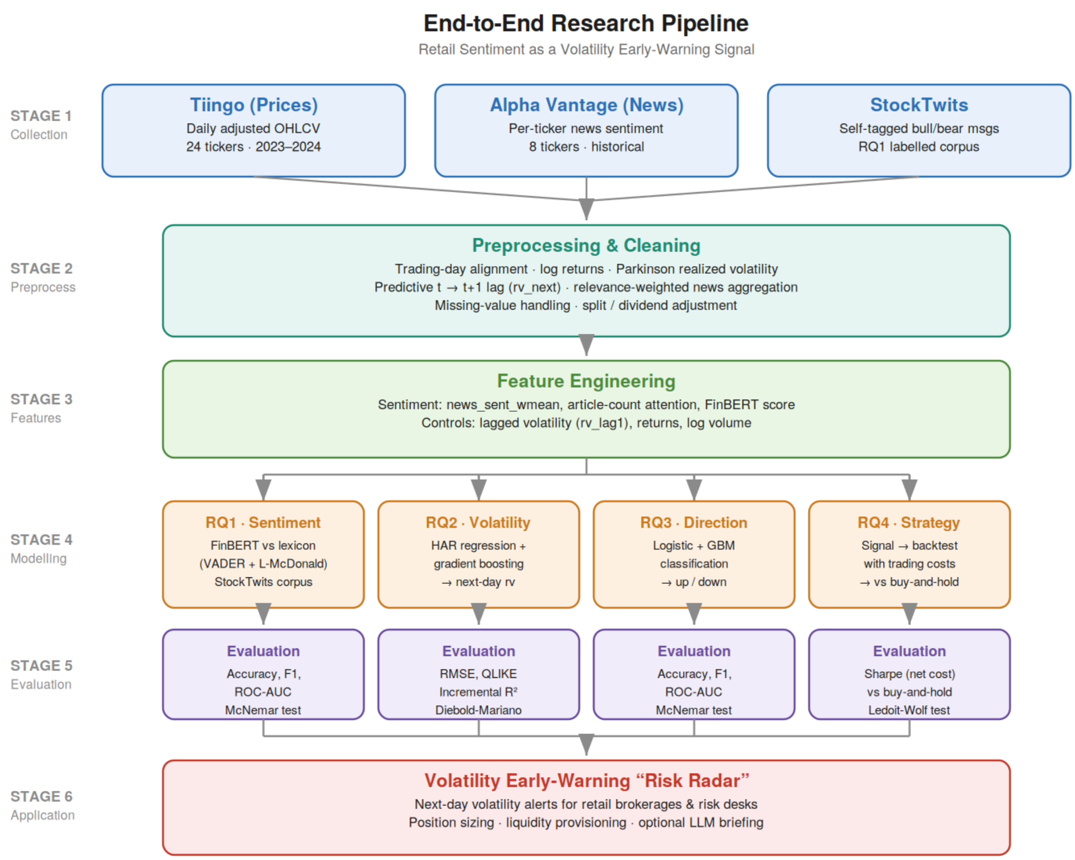
Note. Computed at alpha = .05 and power = .80 with u = 3 tested predictors and L = 10.90 (Cohen, 1988). Cohen's benchmarks are f^2 = 0.02 (small), 0.15 (medium) and 0.35 (large). Generated by src/run_eda.py.

Modelling

The overall research pipeline, from data collection through preprocessing, feature engineering, modelling, and evaluation, is shown in Figure 8.

Figure 8

End-to-End Research Pipeline



Note. The pipeline maps each of the four research questions to its model family and evaluation metrics, with StockTwits feeding RQ1 and Alpha Vantage news feeding RQ2–RQ4.

Data Preprocessing

Preprocessing is deterministic and executed in src/build_features.py, so the panel can be rebuilt from the raw files by one command. Articles are assigned to United States Eastern trading dates, with items published after the 16:00 close attributed to the following trading day, so post-close information informs t + 1. The predictive alignment is created by shifting realized volatility back one trading day within each ticker, defining the target while leaving every predictor dated on or before t.

Missing values are handled by category, because the causes differ. Structural gaps, namely the first 21 trading days of each series where the monthly HAR window is incomplete and the final day where the target is undefined, are removed rather than imputed, since imputing would invent history. Zero-article days are not treated as missing at all, for the reason given in the cleaning subsection. No imputation of prices or volumes was required, the adjusted Tiingo series being complete.

Transformation is dictated by the model. Trading volume and article counts are heavily right-skewed and enter in logarithmic form, and the regression target is log-transformed on the evidence of Figure 1. Predictors are standardised for logistic regression, where a coefficient's scale would otherwise depend on its variable's units; no scaling is applied for gradient boosting, because tree splits are invariant to monotonic rescaling. The only categorical variable is the ticker symbol, absorbed as a fixed effect rather than one-hot encoded, which is the appropriate treatment for a panel where each unit contributes many observations.

Experimental Design and Validation Strategy

Each research question is a controlled comparison. The control condition contains only market information; the experimental condition is the identical model, algorithm and hyperparameters with sentiment features added. Both are estimated on the same training windows and scored on the same test observations, making the comparison paired and isolating sentiment's contribution from any difference in model class. Reproducibility is supported by fixed random seeds, a public repository, and the validation harness described earlier.

Models are validated by walk-forward analysis rather than k-fold cross-validation. Although Step 4 proposes k-fold as the default safeguard against overfitting, random partitioning is not admissible here: assigning observations to folds at random would place future trading days in the training set, allowing the model to learn from information unavailable at the time of the forecast. Walk-forward validation trains on an initial contiguous block, tests on the block that follows, then advances both windows, so every prediction uses only prior data. Splits are formed on dates rather than rows, so all eight tickers move through the windows together.

Regression Diagnostics and Model Selection

The HAR specification is assessed against the assumptions of linear regression. Residuals are inspected for non-linearity and heteroscedasticity, the latter expected in volatility data and addressed by clustering standard errors by date, so that inference accounts for the cross-sectional correlation identified in Figure 5. Normality is examined on the log scale, where Figure 1 shows the target is close to symmetric. Multicollinearity is assessed in Table 5, where no variance inflation factor exceeds the threshold of 10. Nested specifications are compared using adjusted R-squared together with the Akaike and Bayesian information criteria, which penalise the additional parameters the sentiment features introduce.

Choice of Models with Justification

Each research question is framed as a baseline-versus-augmented comparison, so that any improvement is attributable to sentiment rather than to model class. Models were selected on the basis of the literature, the data type, interpretability, and the evaluation task.

Model 1 — FinBERT vs. lexicon (RQ1): a transformer classifier (FinBERT) benchmarked against a VADER + Loughran–McDonald lexicon baseline, graded on StockTwits self-tagged labels. Justified by FinBERT's demonstrated gains in financial sentiment classification (Araci, 2019) and the transparency of the lexicon baseline (Loughran & McDonald, 2011).

Model 2 — HAR regression + gradient boosting (RQ2): a heterogeneous-autoregressive baseline for realized volatility, extended with sentiment features and a gradient-boosting regressor to capture non-linearities. Justified by the HAR model's standing in volatility forecasting and prior sentiment-volatility evidence (Audrino et al., 2020).

Model 3 — Logistic regression + gradient boosting (RQ3): a technicals-only classifier of next-day return direction, compared with a sentiment-augmented version. Justified by interpretability (logistic) and predictive strength (gradient boosting) on tabular data.

Model 4 — Sentiment-informed strategy backtest (RQ4): a transparent trading rule derived from the volatility signal, back-tested with transaction costs against buy-and-hold. Justified by the need to demonstrate economic, cost-aware value rather than statistical significance alone.

Features Included and Feature Engineering

The feature set combines news-derived sentiment features with lagged market controls; the target is the next-day realized volatility. Table 9 summarises the feature set and its usage.

Table 9

Feature Set and Usage

Feature	Type	Engineering	Used in
news_sent_wmean	Sentiment	Relevance-weighted daily mean	RQ2, RQ3, RQ4
news_article_count	Attention	Daily article count (log)	RQ2, RQ3
FinBERT sentiment	Sentiment (DL)	RQ1 model applied to news text	RQ2, RQ3
rv_lag1	Control	Prior-day realized volatility	RQ2
log_return, abs_return	Control	Daily returns	RQ2, RQ3
log_volume	Control	log(1 + volume)	RQ2, RQ3

Evaluation Metrics (Formulae and Calculations)

Metrics are matched to each question's form. Table 10 lists the metrics, their formulae, and the research question to which each applies; worked illustrative calculations are retained from the synopsis and will be recomputed on real results in the final report.

Table 10

Evaluation Metrics and Formulae

Metric	Formula	Applies to
Accuracy	$(TP + TN) / (TP + FP + TN + FN)$	RQ1, RQ3
Precision / Recall	$TP/(TP+FP)$; $TP/(TP+FN)$	RQ1, RQ3
F1	$2(\text{Precision} \times \text{Recall})/(\text{Precision} + \text{Recall})$	RQ1, RQ3
ROC-AUC	Area under TPR-vs-FPR curve	RQ1, RQ3
McNemar's χ^2	$(\lvert b - c\rvert - 1)^2 / (b + c)$	RQ1, RQ3
RMSE	$\sqrt{(1/n)\ \Sigma(\text{pred} - \text{actual})^2}$	RQ2
QLIKE	$(1/n)\ \Sigma[(a/p) - \ln(a/p) - 1]$	RQ2
Sharpe ratio	$(\text{mean} - rf)/sd$, annualised $\times \sqrt{252}$	RQ4

Preliminary Results

Modelling is complete for RQ1 and RQ2 and outstanding for RQ3 and RQ4. Table 11 compares each model against its baseline; hypothesis tests are reported per question below, and all figures are out-of-sample.

RQ1: Sentiment Measurement

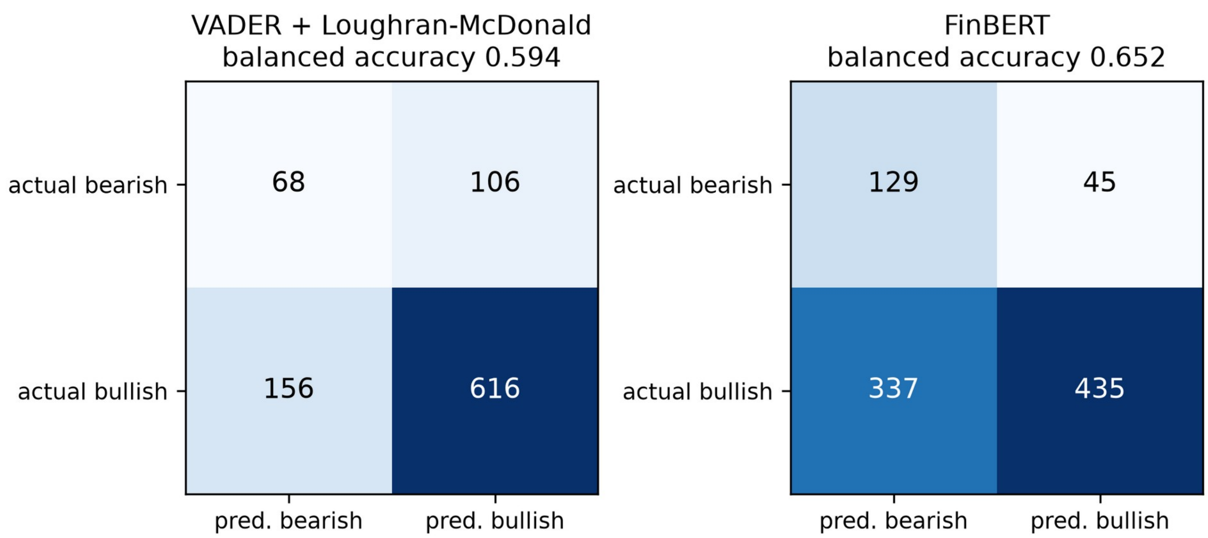
After cleaning, the labelled corpus contains 3,152 messages, 81.6% tagged bullish. The corpus was split 70/30 with stratification and every value below is computed on the held-out 30%. Because both scorers emit a continuous value, the decision threshold was chosen on the training half alone; selecting it on the test data would flatter the models.

The result is more nuanced than the hypothesis anticipated. On raw accuracy FinBERT (0.670) is significantly less accurate than the VADER lexicon (0.739): McNemar's test rejects equality in the lexicon's favour, with 237 messages correct for the lexicon alone against 102 for FinBERT alone (chi-square = 52.97, $p < .001$). Taken at face value this retains the null hypothesis. Accuracy is the wrong criterion on a corpus that is 82% one class, however: answering bullish every time scores 0.816, higher than either model, while carrying no information. The lexicon's advantage has the same source, its calibrated threshold predicting bullish on four-fifths of messages.

On the criteria that account for the class imbalance, FinBERT is ahead. Its balanced accuracy is 0.660 against 0.611, and it identifies 64.4% of bearish messages against 40.8%. Its ranking ability is clearly superior, with an area under the receiver operating characteristic curve of 0.712 against 0.614. A paired bootstrap over 5,000 resamples, permitted by the research design as an alternative to McNemar's test, confirms the ranking advantage is significant (+0.098, 95% CI [+0.053, +0.143], $p < .001$) and that the balanced-accuracy advantage over the combined lexicon is also significant (+0.058, 95% CI [+0.013, +0.103], $p = .012$). The comparable difference against VADER alone is positive but not significant (+0.042, $p = .071$).

Figure 9

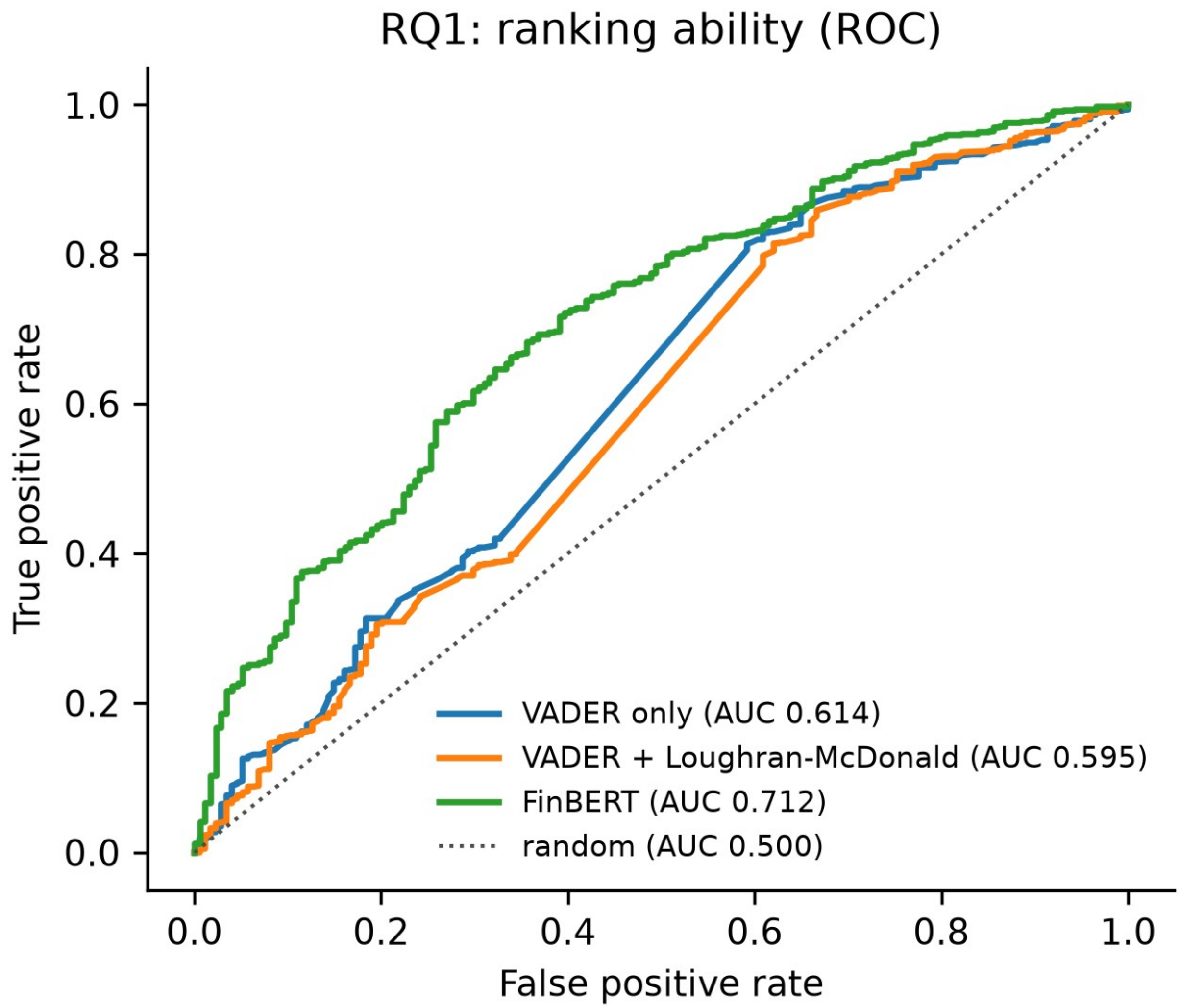
RQ1 Classification Errors by Model



Note. Counts are test-set messages (n = 946). FinBERT recovers 64.4% of bearish messages against the lexicon's 40.8%; the lexicon's higher raw accuracy comes from predicting the bullish majority more often.

Figure 10

RQ1 Ranking Ability (ROC)



Note. Threshold-free comparison. FinBERT's area under the curve of 0.712 exceeds the lexicon's 0.614; the paired bootstrap difference is +0.098, 95% CI [+0.053, +0.143].

RQ2: Volatility Prediction

The RQ2 models were estimated on the full 3,840-observation panel with ticker fixed effects, using expanding-window walk-forward validation across six folds split on dates. The baseline is the heterogeneous autoregressive specification with trading volume; the augmented model adds the three sentiment and attention variables.

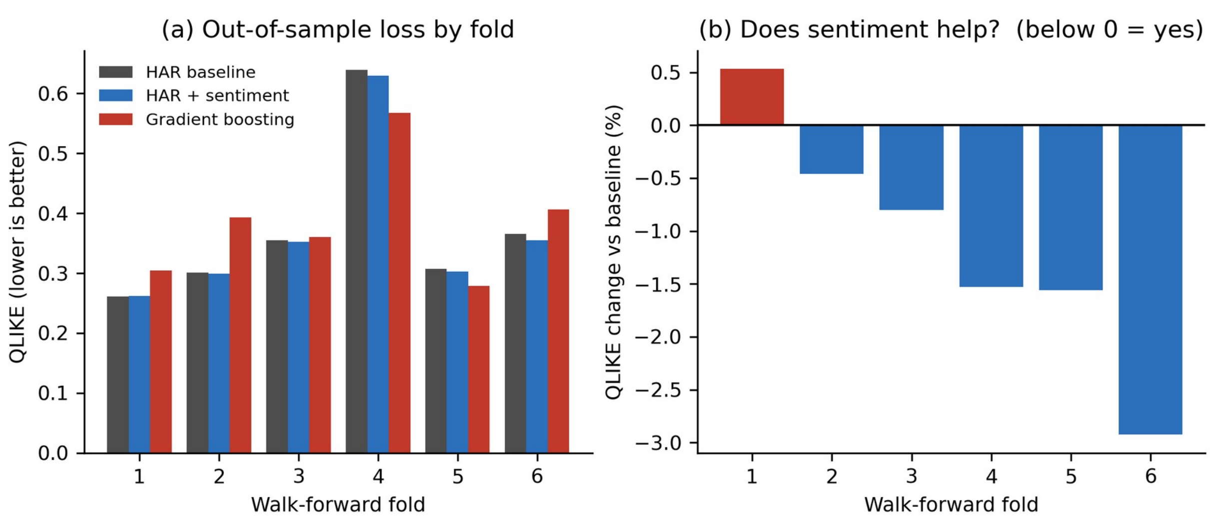
The sentiment block is jointly significant. An incremental F-test rejects the null that its coefficients are all zero ($F = 3.61$, $p = .013$), and adjusted R-squared rises from .5387 to .5401 while both the Akaike and Bayesian criteria fall, so the improvement survives the penalty for three extra parameters. Out-of-sample the augmented model lowers pooled QLIKE by 1.26% and is more accurate in five of six folds; Diebold-Mariano rejects equal predictive accuracy decisively (DM = 4.82, $p < .001$).

The composition of that gain corroborates the exploratory analysis. With standard errors clustered by date, attention carries the effect (coefficient 0.055, $t = 3.12$, $p = .002$) while sentiment tone does not (-0.035, $t = -0.86$, $p = .389$) and the zero-attention indicator is not individually significant ($p = .170$). How much a stock is written about predicts its next-day volatility; whether that coverage reads positive or negative does not. This is the pattern Figure 4 showed before any model was fitted, and it is consistent with Audrino et al. (2020).

Gradient boosting was fitted on the identical feature set and did not improve on the linear specification (DM = -1.25, $p = .211$). With 3,840 observations and a volatility process close to linear in its lagged components, the additional flexibility captures noise rather than structure. The result is reported rather than discarded, since a null finding on model class is itself informative.

Figure 11

RQ2 Out-of-Sample Forecast Loss by Walk-Forward Fold



Note. QLIKE is computed on the variance scale (Patton, 2011); lower is better. Panel (b) expresses the augmented model's loss relative to the baseline, so bars below zero indicate that sentiment helped.

RQ3: Direction Classification

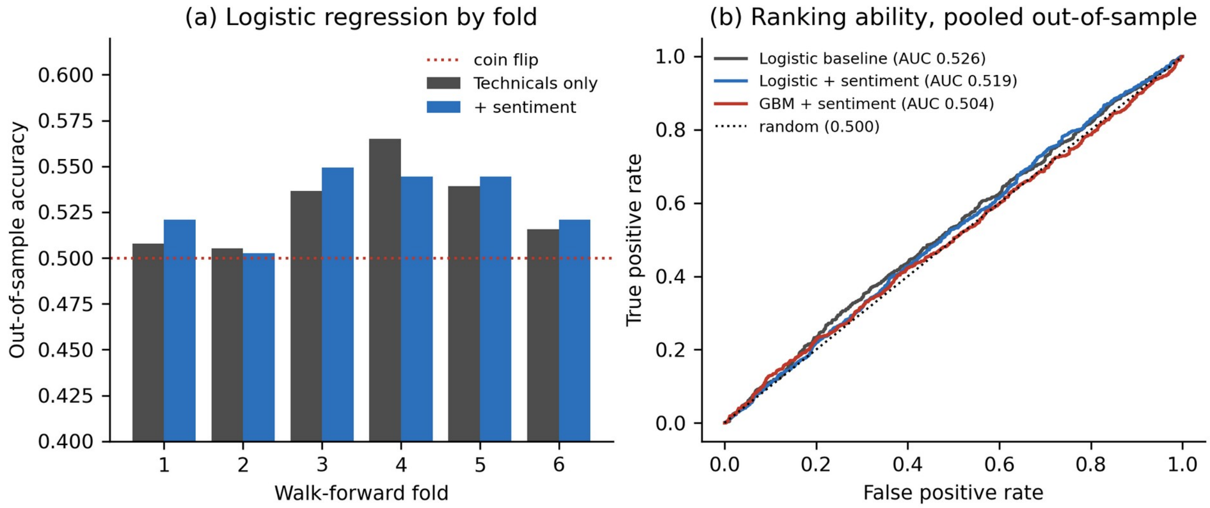
RQ3 reuses the RQ2 panel, ticker fixed effects and walk-forward folds, changing only the target, which becomes the sign of the next day's return, and the model class. Features were standardised for logistic regression using a scaler fitted on each training window alone, and left unscaled for gradient boosting.

Sentiment does not improve direction prediction. Pooled out-of-sample accuracy rises from .5282 to .5304, an increase of 0.22 percentage points that McNemar's test cannot distinguish from chance (chi-square = 0.13, p = .714); the gradient-boosting pair behaves the same way (p = .187). The null hypothesis is retained. The scale of the result is what matters: a constant rule predicting an up day every time scores .5295, so the augmented model beats guessing by nine ten-thousandths, and the area under the curve of .519 is close to the .500 of a coin toss.

This is the outcome the design anticipated, reported as a finding rather than a shortcoming. Read alongside RQ2 the two results form a coherent pair: the same sentiment features that significantly improve next-day volatility forecasts carry no usable information about direction. That asymmetry is the central claim of the literature this study builds on (Antweiler & Frank, 2004; Audrino et al., 2020) and it justifies framing the application as a volatility early-warning signal rather than a return-prediction tool.

Figure 12

RQ3 Direction Prediction Accuracy and Ranking Ability



Note. Panel (a) shows out-of-sample accuracy by fold against the coin-flip line; panel (b) shows pooled receiver operating characteristic curves. All models sit close to random.

Table 11

Preliminary Model Performance

Research Question	Model	Feature Set	Validation Approach	Baseline	Augmented	Key Takeaway
RQ1	FinBERT vs VADER lexicon	Message text	Stratified 70/30 split; threshold fitted on train	AUC .614	AUC .712	FinBERT ranks significantly better (+.098, p < .001) but is less accurate on a 82/18 corpus
RQ1	FinBERT vs VADER lexicon	Message text	As above	Bal. acc .611	Bal. acc .660	Advantage significant against the combined lexicon (p = .012)
RQ2	HAR + sentiment (OLS)	rv, rv_w, rv_m, log_volume, ticker FE, + 3 sentiment variables	Expanding-window walk-forward, 6 folds on dates	QLIKE .3715	QLIKE .3668	Sentiment block jointly significant (F = 3.61, p = .013); DM = 4.82, p < .001
RQ2	Gradient boosting	As above	As above	QLIKE .3715	QLIKE .3861	No improvement over linear HAR (DM = -1.25, p =

						.211)
RQ3	Logistic + gradient boosting	As RQ2; target = next-day return direction	As RQ2	Acc .5282	Acc .5304	No improvement; McNemar chi-square = 0.13, p = .714. Direction is near-random, as anticipated
RQ4	Strategy vs buy-and-hold	RQ2 volatility forecast	Cost-aware backtest; block-bootstrap Ledoit-Wolf	pending	pending	Power-constrained; see limitations

Note. All metrics are out-of-sample. QLIKE is computed on the variance scale (Patton, 2011); lower is better. AUC denotes the area under the receiver operating characteristic curve. Generated by src/run_rq1.py and src/run_rq2.py.

Link to the Research Questions

RQ1 confirms the sentiment instrument is sound: FinBERT separates bullish from bearish messages better than a lexicon, though not more accurately in a raw sense. RQ2 answers the central question affirmatively, with attention rather than tone supplying the predictive content, which narrows what the risk-radar signal should be built on. RQ3 returns a null result that corroborates rather than undermines RQ2, confirming the volatility focus. RQ4 alone remains outstanding, converting the volatility forecast into an exposure rule under the power limitation already noted.

Interim Limitations and Risks

- Free-tier data quota (Alpha Vantage, 25 requests/day) constrains the sentiment universe to eight tickers and slows collection; mitigated by caching and resumable, per-ticker pulls.
- StockTwits serves only recent messages, so it supports RQ1 validation rather than the historical panel; mitigated by using Alpha Vantage news as the historical sentiment feature.
- Sentiment effects are expected to be economically small (per the literature); mitigated by conservative effect-size assumptions in the sample-size design.

- Rural hotspot connectivity causes intermittent collection timeouts; mitigated by retry/backoff and the option to collect from a stable cloud environment.

Statistical power for RQ4 is limited: the strategy test requires 1,558 trading days and the study window provides 500, so the backtest is under-powered and a null result there is inconclusive rather than negative; mitigated by a block-bootstrap implementation of the Ledoit-Wolf test and by explicit reporting of the shortfall.

Effective sample size is materially smaller than the row count. The eight tickers move together, with a mean pairwise correlation of 0.359 in realized volatility, so the 3,840 ticker-days correspond to roughly 2.3 independent series and an effective sample nearer 1,093 observations; mitigated by reporting standard errors clustered by date and by sizing claims against the adjusted figure rather than the raw row count.

The study window contains a single volatility regime. Quarterly cross-sectional mean volatility varies only 1.25-fold across 2023-2024, with no crisis or sustained stress episode, so the models are trained and validated under calm to moderate conditions only; mitigated by restricting all claims to comparable conditions and identifying regime generalisation as future work.

Attention is measured from news-media coverage rather than retail activity. The direct retail measure would be StockTwits message volume, which is unavailable historically because the API serves only recent messages; mitigated by treating article count as a proxy and stating the substitution explicitly.

Next Steps (for Final Report)

- Complete Alpha Vantage news collection for all eight sentiment tickers.
- Assemble the final analysis panel and run data cleaning and EDA with interpretation.
- Train and validate FinBERT against the lexicon baseline (RQ1).

- Fit and evaluate the volatility (RQ2) and direction (RQ3) models out-of-sample.
- Run the cost-aware strategy backtest and significance test (RQ4).
- Report preliminary and final results, and draft the implementation, recommendation, and limitations sections.

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Appendix

[Optional. Include any code referenced in the text (e.g., feature-building or collection scripts), with each appendix titled and referenced from the body.]